

Review: The real number system



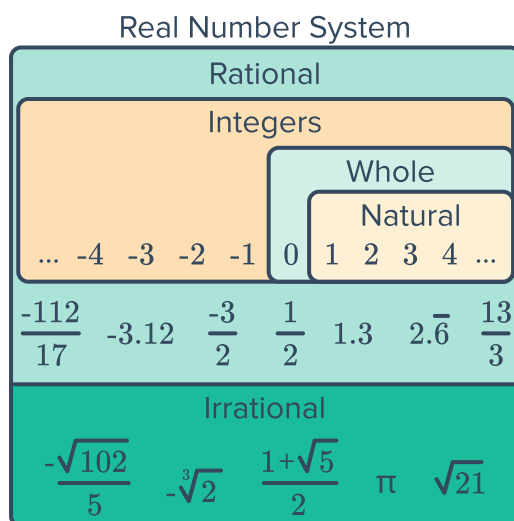
1 Consider the following sets of numbers.

- $\{0, 1, 2, 3, \dots\}$
- $\{\dots, -3, -2, -1, 0, 1, 2, 3, \dots\}$
- $\{1, 2, 3, \dots\}$
- $\{\dots, -3, -2, -1, 0\}$

- a Identify the set of all integers.
- b Identify the set of whole numbers.
- c Identify the smallest whole number.

2 Classify the set of numbers containing $\{0, 1, 2, \dots\}$

3 Consider the following diagram:



a Determine whether each of the following statements is true or false.

- i $\sqrt{13}$ is a rational number.
- ii When written in decimal form, irrational numbers have a repeating pattern of decimal digits.
- iii A real number is either a rational or irrational number.

b Classify the following numbers:

- | | | | |
|---------------|--------------------|----------------|----------------------|
| i 38 | ii -42 | iii $\sqrt{6}$ | iv 3.058 |
| v $\sqrt{49}$ | vi $\frac{\pi}{3}$ | vii 0.5777... | viii $\frac{26}{45}$ |



- 4 Every whole number is also a rational number, but not every rational number is a whole number. Explain why this statement is true.
- 5 Christa decides to express her favorite rational number as a decimal. Describe this decimal.
- 6 Classify the set of number most appropriate to describe the following:
- a The populations of towns
 - b The times of the runners (in seconds) in a 100 m sprint.
 - c Distance between Jupiter and Venus.
 - d The position of a submarine relative to sea level. (Note that: height above sea level is expressed as a positive quantity.)
 - e The goal difference of a hockey team at the end of a season. (Note that: The goal difference is the number of goals the team scored in the season minus the number of goals that were scored against them.)
 - f The cost (in dollars) of sending a text message.
- 7 The circumference of a circle is given by $2\pi r$, where r is the radius. State whether each of the following statements is true about the circumference of the circle.
- a The circumference is an irrational number if the radius is rational.
 - b The circumference is a rational number if the radius is equal to π .
 - c The circumference is a rational number if the radius is irrational.
 - d The circumference is a rational number if the radius is equal to $\frac{a}{\pi}$ where a is an integer.

Additional questions

- 8 Give a definition for the subset of rational numbers.
- 9 Give a definition for the subset of irrational numbers.
- 10 Can a number be both rational and irrational? Explain.
- 11 Do integers belong to the subset of rational or irrational numbers? Explain.



- 12** The following numbers are given in their simplest form.
For each number, state whether it is rational or irrational.

a $-\frac{2}{7}$

b $\sqrt{2}$

c 8

d 20.57

e $3\sqrt{5}$

f $1 - \sqrt{7}$

g $-0.\bar{3}$

h π

- 13** The following numbers are not given in their simplest form.
Simplify each number and state whether it is rational or irrational.

a $\sqrt{16}$

b $\frac{2\pi}{8}$

c $-\frac{4\pi}{5\pi}$

d $\frac{0.4}{5}$

- 14** Akin states that $24.737337333\dots$ is a rational number because it has a repeating pattern.
Is Akin correct? Explain.